

Suyog Neupane

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Education

The University of Texas at Arlington

Bachelor of Science in Electrical Engineering (Honors), Minor in Computer Science

GPA: 3.67/4.0

Relevant Coursework: Circuit Analysis II, Electronics I, Digital Circuits, Control Systems

Aug 2023 to May 2027

Arlington, TX

Skills

Programming Languages: C, C++, Python, SystemVerilog

Frameworks & Libraries: MATLAB/Simulink, ROS2, NAV2, CUDA

Tools & Software: Circuit and Schematic Design, Multisim, SOLIDWORKS, NI LabVIEW, Jetson Nano, Git

Experience

University of Texas at Arlington Research Institute (UTARI)

Research Assistant

Feb 2024 to Present

Arlington, TX

- Engineered soft robotic systems for upper body motion assistance and rehabilitation.
- Developed embedded software in C++ for precise finger, wrist, and hand control with roughly 95% tracking accuracy.
- Designed, modeled, and tested soft pneumatic wrist exoskeletons using MATLAB/Simulink, achieving 55° of wrist range of motion across multiple operating conditions.
- Built embedded control for a MEG compatible vibrotactile therapy glove, delivering frequency accurate stimulation (measured 20.0 Hz and 30.5 Hz) with zero magnetic artifacts across 2 pediatric participants.

AutonoBots UTA

Robotics Engineer

Feb 2024 to Jan 2025

Arlington, TX

- Built a navigation system for The Boring Company competition using ROS2, NAV2, and Rviz on a Clearpath Jackal.
- Accelerated YOLO model inference on a Jetson Nano using CUDA and distributed computing for 3x faster response.

Squad of Change Makers

Electrical Engineer

2022 to 2023

Nepal

- Owned the full avionics stack, hardware to firmware, for Nepal's first amateur rocket, built at age 17.
- Integrated IMU and barometric altitude tracking with onboard SD data logging across the full flight profile.
- Programmed automatic parachute deployment triggered by freefall detection for reliable payload recovery.

Publications

- A Novel Design of Two DoFs Soft Pneumatic Wrist Exoskeleton for Rehabilitation.
- Experimental Data Driven Angular Velocity Control for a Soft Pneumatic Wrist Exoskeleton for Motion Assistance in Rehabilitation.
- Learning based Approach to Assist Occupational Workers using Soft Pneumatic Wrist Exoskeleton.

Projects

Haptic Focus Assistant | *Embedded C++, BLE*

Jan 2026 to May 2026

- Hardware lead on a BLE connected wrist wearable that buzzes on distracting app use; owned the full circuit schematic, embedded algorithm, power regulation, and motor driver stage.
- Led a team of 3 and cut average daily screen time from roughly 8 hours to 3 hours across three user studies.

FPGA Options Strategy Engine | *SystemVerilog, FPGA*

Jan 2026 to Present

- Built a SystemVerilog pipeline evaluating multiple leg options strategies on market data, targeting tick to trade latency in the tens of nanoseconds, roughly two orders of magnitude below an equivalent software path.
- Verified in simulation against a Python golden model on synthetic and recorded feed data.

Mars Autonomous Rover Rally | *Python*

Nov 2025 to Jan 2026

- Team lead for UT Arlington's Space Teams University entry; built the Python AI Mission Manager driving the rover between waypoints on simulated Martian terrain with camera based rock detection and evasive steering. Placed 4th among university teams from four countries.

Awards

- Genspiration Prize (\$5000) for the control algorithm of a novel 2 DoF wrist exoskeleton.
- Placed 4th internationally among university teams at the Space Teams University competition winning \$2500.